

Large Language Diffusion Models

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Is Autoregression Necessary for LLM Capabilities?

- Generative modeling principles

$$\max_{\theta} \mathbb{E}_{p_{\text{data}}(x)} \log p_{\theta}(x) \Leftrightarrow \min_{\theta} \text{KL}(p_{\text{data}}(x) || p_{\theta}(x)) .$$

- Autoregressive formulation

$$p_{\theta}(x) = p_{\theta}(x^1) \prod_{i=2}^L p_{\theta}(x^i \mid x^1, \dots, x^{i-1})$$

Is Autoregression Necessary for LLM Capabilities?

- LLMs are generative models; common practice is autoregressive factorization (left-to-right).
- Core question: do scalability, in-context learning, instruction-following require autoregression?
- Hypothesis: it is generative modeling principles (likelihood-based learning) that matter, not the AR factorization itself.
- Autoregression induces limitations (e.g., reversal reasoning failures).

Headline Results

- LLaDA scales to 8B from scratch and is competitive with strong AR LLMs on diverse benchmarks.
- Shows strong in-context learning; after SFT, strong instruction-following in case studies.
- Addresses the “reversal curse” with more balanced forward vs reversal performance

Headline Results

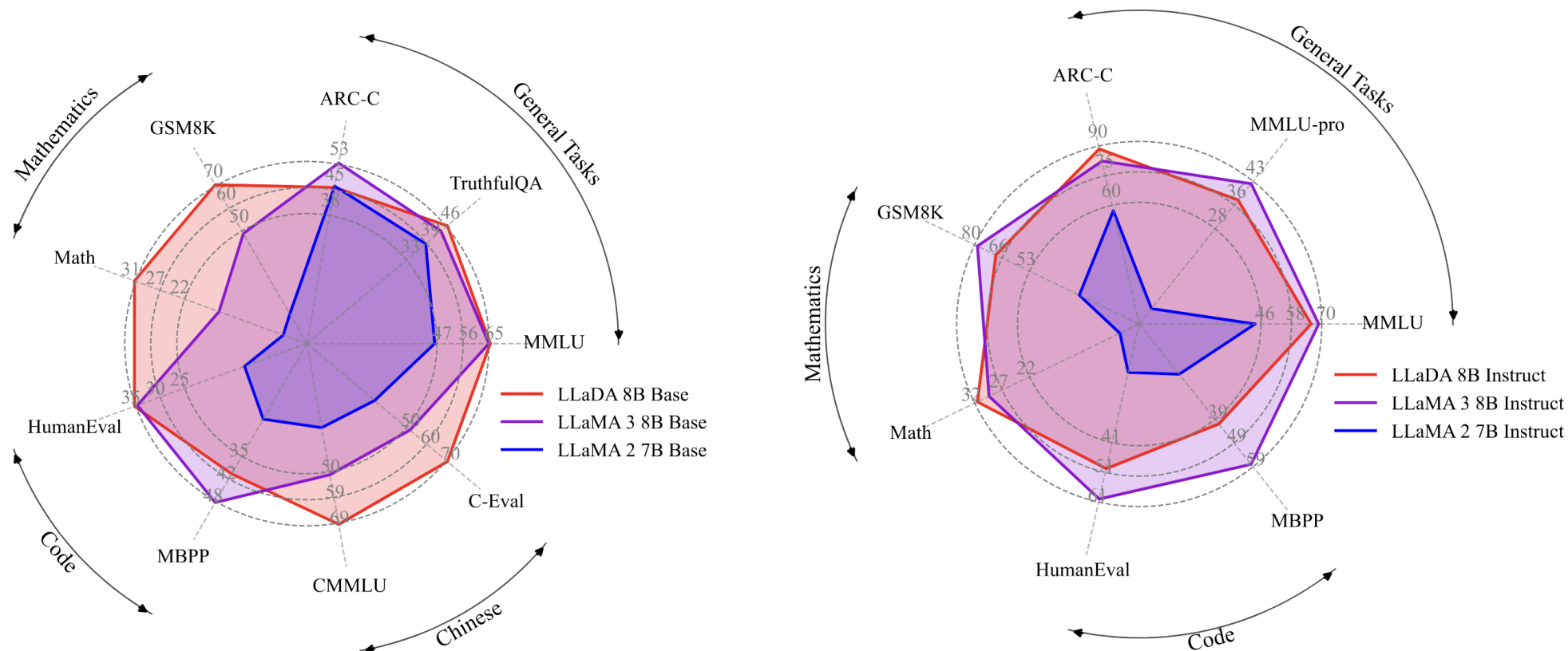


Figure 1: **Zero/Few-Shot Benchmarks.** We scale LLaDA to 8B parameters from scratch and observe competitive zero/few-shot performance compared with strong autoregressive LLMs [6].

Overview of LLaDA

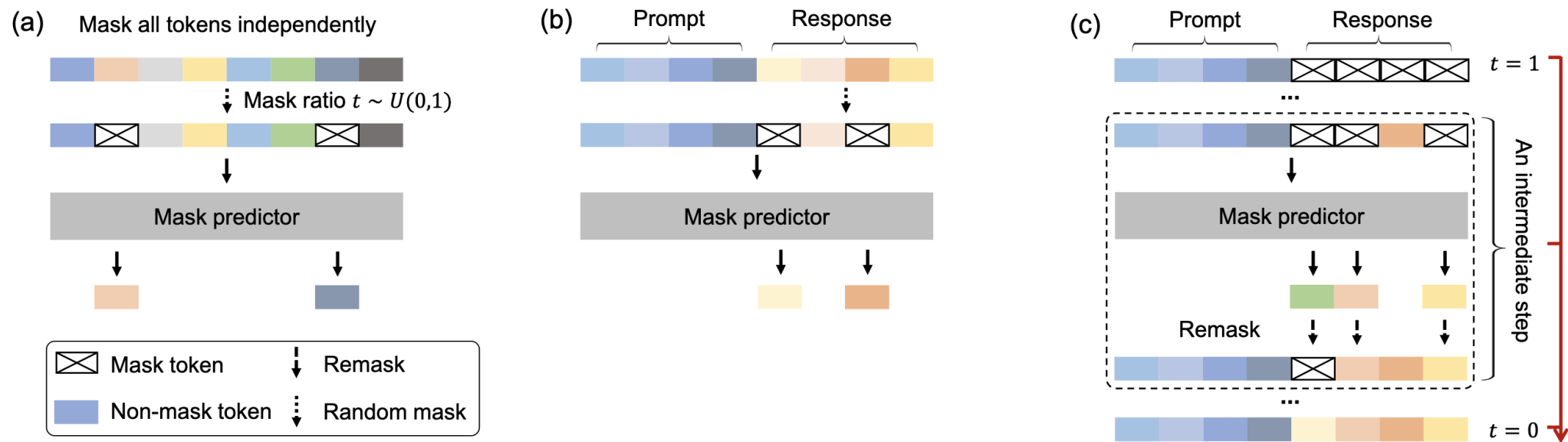


Figure 2: **Overview of LLaDA.** (a) Pre-training. LLaDA is trained on text with random masks applied independently to all tokens at the same ratio $t \sim U[0, 1]$. (b) SFT. Only response tokens are possibly masked. (c) Sampling. LLaDA simulates a diffusion process from $t = 1$ (fully masked) to $t = 0$ (unmasked), predicting all masks simultaneously at each step with flexible remask strategies.

- LLaDA = masked diffusion model for language (discrete diffusion via masking).
- Emphasis: bidirectional dependencies (no causal mask), likelihood-based objective.

Overview of LLaDA



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Generative Modeling Setup

- Data x_0 (token sequence, length L).
- Forward process: independently mask each token with probability t to form x_t , where $t \sim \mathcal{U}[0,1]$.
- Reverse process: a mask predictor $p_\theta(\cdot | x_t)$ predicts **all masked tokens simultaneously**.
- Contrast with BERT: fixed mask ratio vs LLaDA's continuously randomized masking ratio.

Training Objective

- Objective: cross-entropy computed **only on masked tokens**, with a $1/t$ weighting.

$$\mathcal{L}(\theta) \triangleq -\mathbb{E}_{t, x_0, x_t} \left[\frac{1}{t} \sum_{i=1}^L \mathbf{1}[x_t^i = \mathbf{M}] \log p_{\theta}(x_0^i | x_t) \right]$$

- The loss upper-bounds the negative log-likelihood of the induced model distribution.

$$-\mathbb{E}_{p_{\text{data}}(x_0)} [\log p_{\theta}(x_0)] \leq \mathcal{L}(\theta)$$

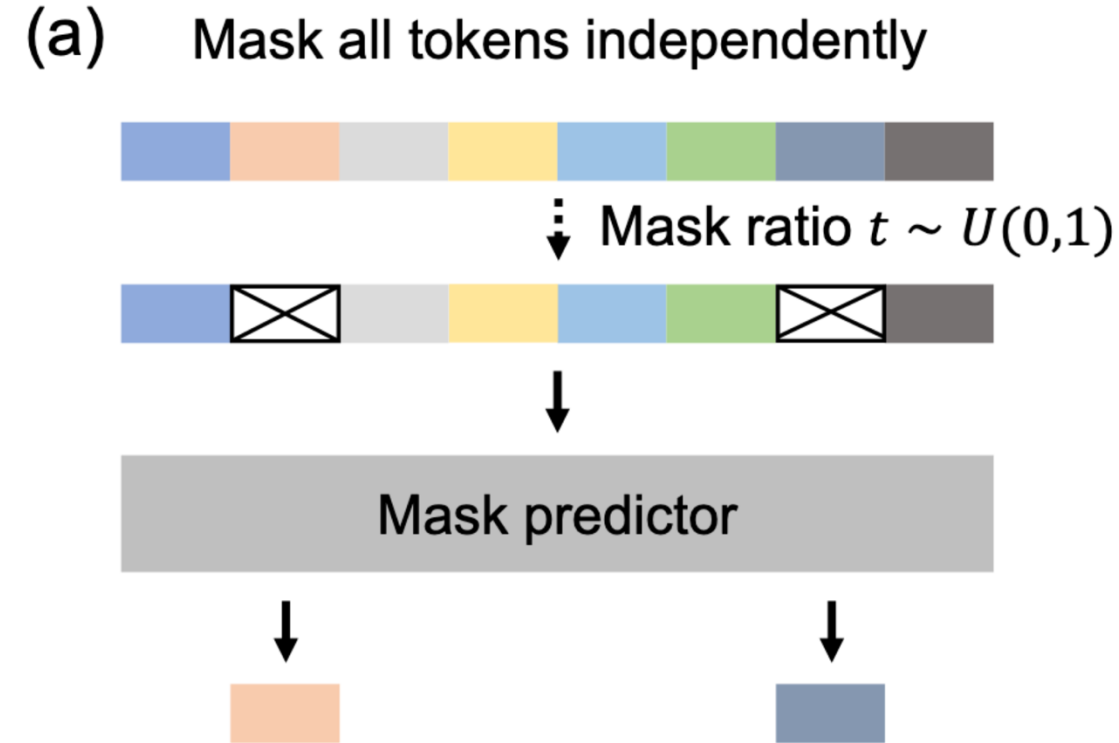
- Practical implication: principled generative modeling at scale (vs heuristic masked modeling).

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Pre-training: Architecture and Key Design Choices

- Transformer mask predictor (no causal mask; sees full context).
- LLaDA is incompatible with KV caching $\Rightarrow$ design differences vs LLaMA-style AR models.
- Two model sizes discussed: 1B and 8B.



Pre-training: Algorithm

Algorithm 1 Pre-training of LLaDA

Require: mask predictor p_θ , data distribution p_{data}

1: repeat

2: $x_0 \sim p_{\text{data}}$ # with a probability of 1%, the sequence length of x_0 follows $\text{U}[1, 4096]$

3: $t \sim \text{U}(0, 1]$

4: $x_t \sim q_{t|0}(x_t|x_0)$ # $q_{t|0}$ is defined in Eq. (7)

5: Calculate $\mathcal{L} = -\frac{1}{t_{*L}} \sum_{i=1}^L \mathbf{1}[x_t^i = \mathbf{M}] \log p_\theta(x_0^i | x_t)$ # L is the sequence length of x_0

6: Calculate $\nabla_{\theta} \mathcal{L}$ and run optimizer.

7: until Converged

8: Return p_θ

Pre-training Data and Compute

- Dataset: 2.3T tokens (general + code + math + multilingual), filtered for quality.
- Sequence length: 4096; includes a small portion of variable-length training (random length) to improve robustness.
- Compute: training cost reported in H800 GPU-hours.
- Learning rate schedule: Warmup-Stable-Decay
 - 1.2T tokens: linearly increased the learning rate from 0 to 4×10^{-4} over the first 2000 iterations
 - 0.8T tokens: decayed the learning rate to 1×10^{-4} and held it constant to ensure stable training
 - 0.3T tokens: linearly reduced the learning rate from 1×10^{-4} to 1×10^{-5}
- Optimizer: AdamW (weight decay of 0.1)
- Batch Size: 1280

Model Architecture Comparison

- Highlight how LLaDA compares to the constructed AR baselines and to LLaMA3 8B.
- Call out: layers/width/heads, vocab size differences, KV-head differences, parameter counts.

Table 5: Model Architecture. We report the architectural configurations for our 1B and 7B ARM baselines, the 1B and 8B LLaDA models, and the 8B LLaMA3 model.

	Our ARM Baseline 1B	LLaDA 1B	Our ARM Baseline 7B	LLaDA 8B	LLaMA3 8B
Layers	22	22	28	32	32
Model dimension	2048	2048	4096	4096	4096
Attention heads	32	32	32	32	32
Vocabulary size	126,464	126,464	126,464	126,464	128,000
FFN dimension	5634	5634	13,440	12,288	14,336
Key/Value heads	4	4	8	32	8
Total parameters	1.49 B	1.49 B	6.83 B	8.02 B	8.03 B
Non-embedding parameters	0.97 B	0.97 B	5.80 B	6.98 B	6.98 B

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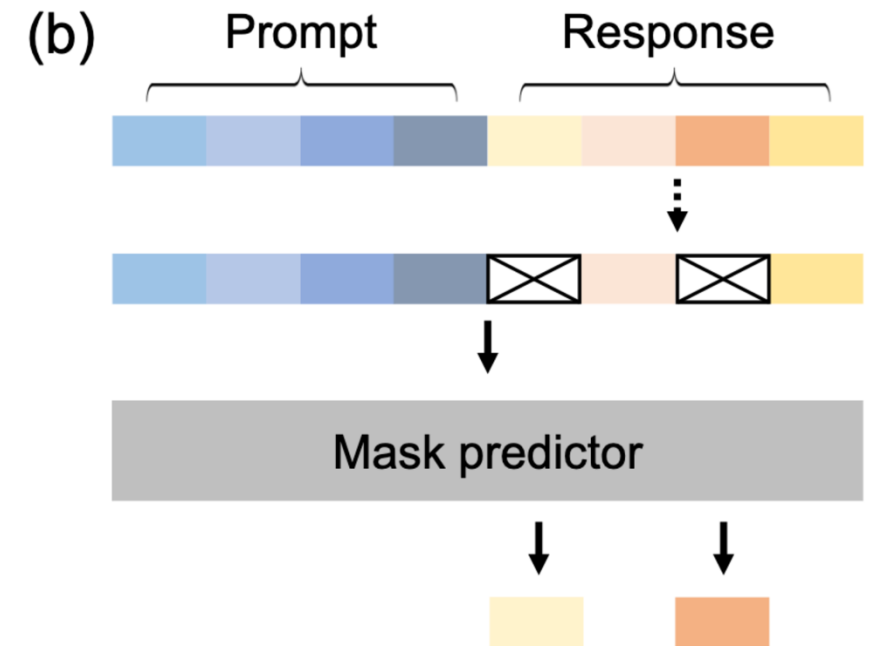
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Supervised Fine-Tuning (SFT) for Instruction Following

- Model the conditional distribution $p_{\theta}(r_0 | p_0)$ (prompt p_0 , response r_0).
- Masking: keep prompt unmasked; only response tokens can be masked at ratio t .
- Loss computed only on masked response tokens.

$$-\mathbb{E}_{t,p_0,r_0,r_t} \left[\frac{1}{t} \sum_{i=1}^{L'} \mathbf{1}[r_t^i = \mathbf{M}] \log p_{\theta}(r_0^i | p_0, r_t) \right]$$

- Note: SFT data handling (EOS padding + multi-turn partitioning) affects sampling behavior later.



Supervised Fine-Tuning (SFT): Algorithm

Algorithm 2 Supervised Fine-Tuning of LLaDA

Require: mask predictor p_θ , pair data distribution p_{data}

1: **repeat**

2: $p_0, r_0 \sim p_{\text{data}}$

please refer to Appendix B.1 for details about the SFT dat

3: $t \sim \text{U}(0, 1]$

4: $r_t \sim q_{t|0}(r_t|r_0)$

$q_{t|0}$ is defined in Eq. (7)

5: Calculate $\mathcal{L} = -\frac{1}{t * L'} \sum_{i=1}^{L'} \mathbf{1}[r_t^i = \text{M}] \log p_\theta(r_0^i | p_0, r_t)$

L' is the sequence length of r_0

6: Calculate $\nabla_\theta \mathcal{L}$ and run optimizer.

7: **until** Converged

8: **Return** p_θ

Supervised Fine-Tuning (SFT) for Instruction Following

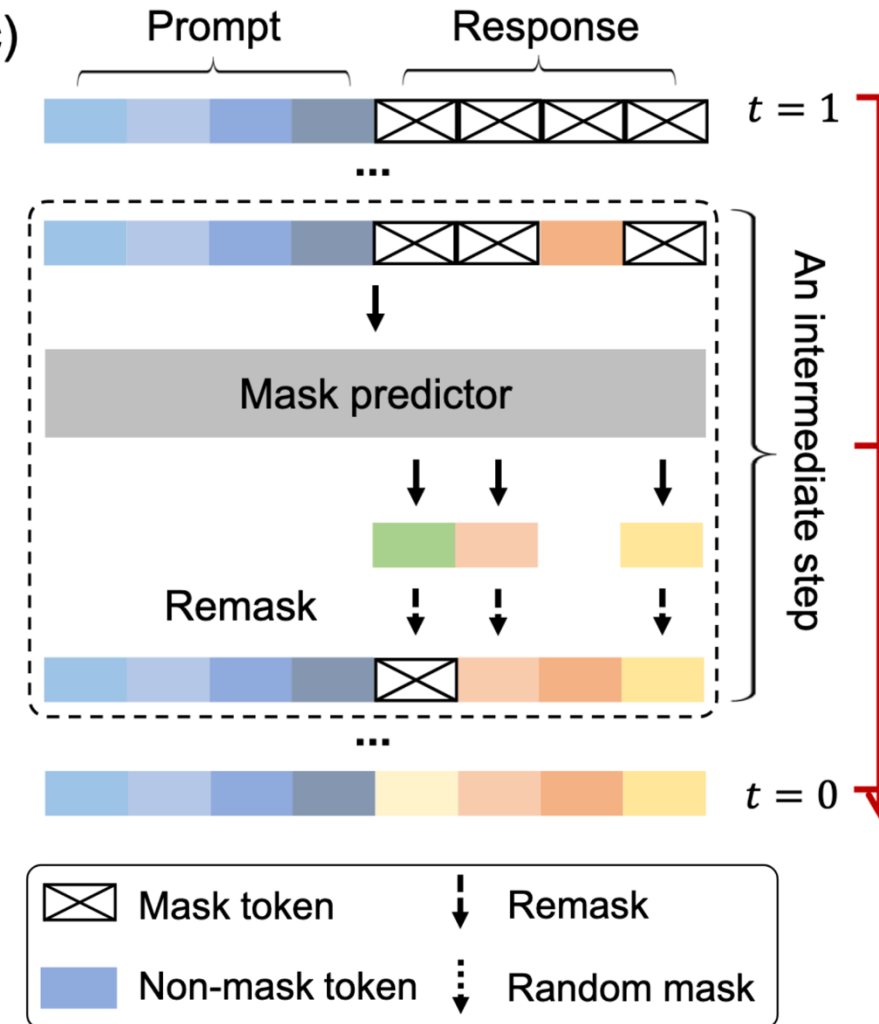
- LLaDA 8B model undergoes SFT on a dataset comprising 4.5 million pairs.
- Append |EOS| tokens to the end of short pairs.
- Treat |EOS| as a normal token during training and remove it during sampling, enabling LLaDA to control the response length automatically.
- Epochs: 3
- Learning Rate
 - 50 iterations + more: linearly increased from 0 to 2.5×10^{-5} and kept constant.
 - final 10% of iterations: linearly reduced to 2.5×10^{-6} .
- Batch Size: 256

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Inference: Diffusion-Style Sampling with Remasking

- Start from fully masked sequence ($t = 1$) and iteratively (c) reduce mask ratio to $t = 0$.
- Each step: predict masked tokens in parallel; optionally remask some tokens for refinement.
- Remasking strategies discussed : random vs low-confidence.



Inference: Algorithm

Algorithm 3 Conditional Log-likelihood Evaluation of LLaDA

Require: mask predictor p_θ , prompt p_0 , response r_0 , the number of Monte Carlo estimations n_{mc}

- 1: $\text{log_likelihood} = 0$
- 2: **for** $i \leftarrow 1$ to n_{mc} **do**
- 3: $l \sim \{1, 2, \dots, L\}$ # L is the sequence length of r_0
- 4: Obtain r_l by uniformly sampling l tokens from r_0 without replacement for masking
- 5: $\text{log_likelihood} = \text{log_likelihood} + \frac{L}{l} \sum_{i=1}^L \mathbf{1}[r_l^i = \mathbf{M}] \log p_\theta(r_0^i | p_0, r_l)$
- 6: **end for**
- 7: $\text{log_likelihood} = \text{log_likelihood} / n_{mc}$
- 8: **Return** log_likelihood

Inference: Algorithm

Algorithm 4 Random Remasking Strategy of LLaDA

Require: mask predictor p_θ , prompt p_0 , answer length L , sampling steps N

- 1: Set r_1 is a fully masked sequence of length L .
 - 2: **for** $t \leftarrow 1$ **down to** $\frac{1}{N}$ **step** $\frac{1}{N}$ **do**
 - 3: $s = t - \frac{1}{N}$
 - 4: $r_0 = \arg \max_{r_0} p_\theta(r_0 | p_0, r_t)$ # we employ greedy sampling when predicting masked tokens
 - 5: **for** $i \leftarrow 1$ to L **do**
 - 6: **if** $r_t^i \neq \text{M}$ **then**
 - 7: $r_0^i = r_t^i$
 - 8: **else**
 - 9: with probability $\frac{s}{t}$, r_0^i is set to M
 - 10: **end if**
 - 11: **end for**
 - 12: $r_s = r_0$
 - 13: **end for**
 - 14: **Return** r_0
-

Inference: Algorithm

Algorithm 5 Low-confidence Remasking Strategy of LLaDA

Require: mask predictor p_θ , prompt p_0 , answer length L , sampling steps N

```
1: Set  $r_1$  is a fully masked sequence of length  $L$ .
2: for  $t \leftarrow 1$  down to  $\frac{1}{N}$  step  $\frac{1}{N}$  do
3:    $s = t - \frac{1}{N}$ 
4:   for  $i \leftarrow 1$  to  $L$  do
5:     if  $r_t^i \neq \text{M}$  then
6:        $r_0^i = r_t^i, c^i = 1$ 
7:     else
8:        $r_0^i = \arg \max_{r_0^i} p_\theta(r_0^i | p_0, r_t)$ 
9:        $c^i = p_\theta(r_0^i | p_0, r_t)_{r_0^i}$ 
10:    end if
11:  end for
12:   $n_{un} = \lfloor L(1 - s) \rfloor$  # the number of unmasked tokens is  $n_{un}$  in timestep  $s$ 
13:  for  $i \leftarrow 1$  to  $L$  do
14:    if  $c^i \in \text{Lowest} - n_{un}(\{c^i\}_1^L)$  then
15:       $r_0^i = \text{M}$  # the  $n_{un}$  positions with the least confidence are selected for remasking.
16:    end if
17:  end for
18:   $r_s = r_0$ 
19: end for
20: Return  $r_0$ 
```

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Scalability Results Across Compute

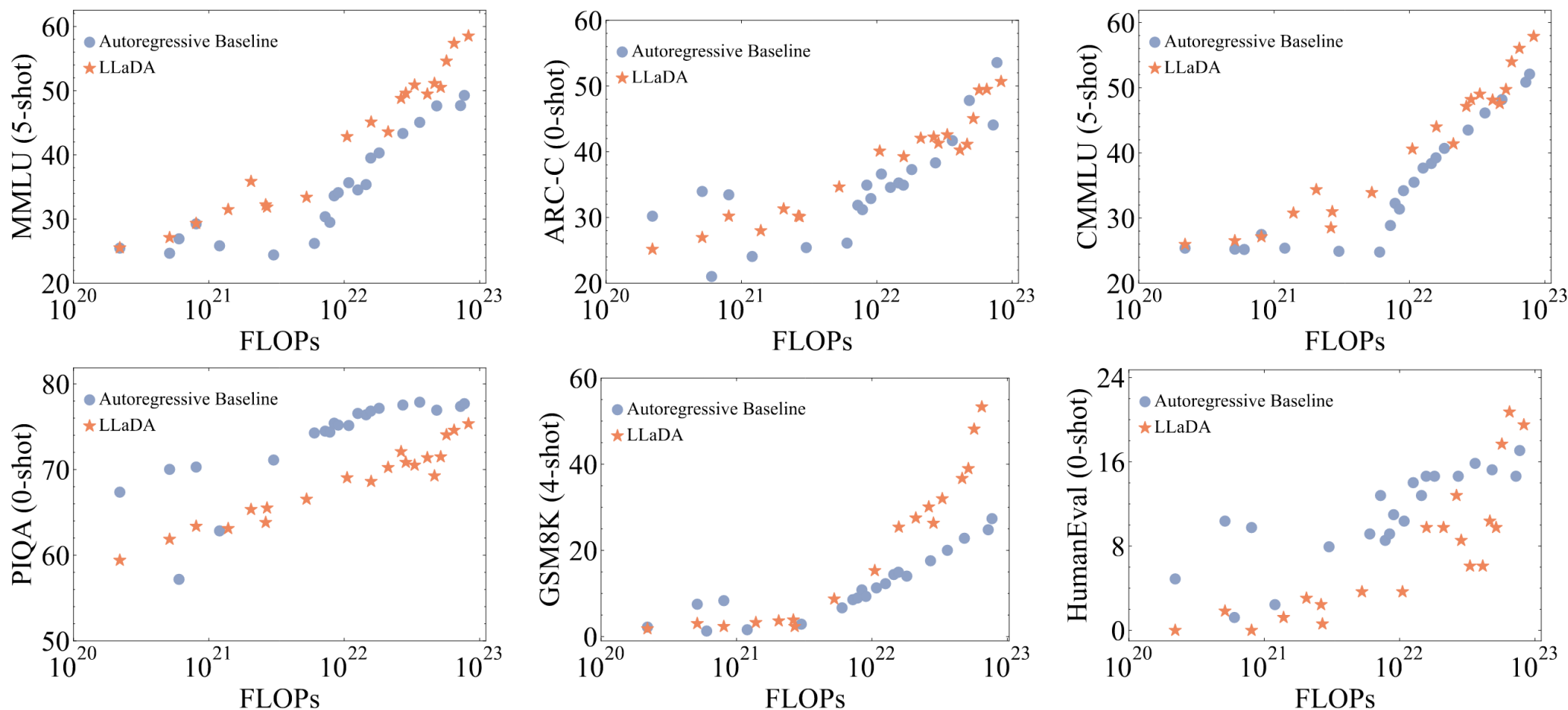


Figure 3: Scalability of LLaDA. We evaluate the performance of LLaDA and our ARM baselines trained on the same data across increasing pre-training computational FLOPs. LLaDA exhibits strong scalability, matching the overall performance of ARMs on six tasks.



Scalability Results Across Compute

- Compare LLaDA vs AR baselines trained on the same data, across increasing pre-training FLOPs.
- Show that LLaDA scales strongly and matches ARMs on multiple tasks.

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Pre-trained Benchmark Results (In-Context Learning)

Table 1: **Benchmark Results of Pre-trained LLMs.** * indicates that models are evaluated under the same protocol, detailed in Appendix B.6. Results indicated by [†] and [¶] are sourced from Yang et al. [25, 26] and Bi et al. [32] respectively. The numbers in parentheses represent the number of shots used for in-context learning. “-” indicates unknown data.

	LLaDA 8B*	LLaMA3 8B*	LLaMA2 7B*	Qwen2 7B [†]	Qwen2.5 7B [†]	Mistral 7B [†]	Deepseek 7B [¶]
Model Training tokens	Diffusion 2.3T	AR 15T	AR 2T	AR 7T	AR 18T	AR -	AR 2T
General Tasks							
MMLU	65.9 (5)	65.4 (5)	45.9 (5)	70.3 (5)	74.2 (5)	64.2 (5)	48.2 (5)
BBH	49.7 (3)	62.1 (3)	39.4 (3)	62.3 (3)	70.4 (3)	56.1 (3)	39.5 (3)
ARC-C	45.9 (0)	53.1 (0)	46.3 (0)	60.6 (25)	63.7 (25)	60.0 (25)	48.1 (0)
Hellaswag	70.5 (0)	79.1 (0)	76.0 (0)	80.7 (10)	80.2 (10)	83.3 (10)	75.4 (0)
TruthfulQA	46.1 (0)	44.0 (0)	39.0 (0)	54.2 (0)	56.4 (0)	42.2 (0)	-
WinoGrande	74.8 (5)	77.3 (5)	72.5 (5)	77.0 (5)	75.9 (5)	78.4 (5)	70.5 (0)
PIQA	73.6 (0)	80.6 (0)	79.1 (0)	-	-	-	79.2 (0)
Mathematics & Science							
GSM8K	70.3 (4)	48.7 (4)	13.1 (4)	80.2 (4)	85.4 (4)	36.2 (4)	17.4 (8)
Math	31.4 (4)	16.0 (4)	4.3 (4)	43.5 (4)	49.8 (4)	10.2 (4)	6.0 (4)
GPQA	25.2 (5)	25.9 (5)	25.7 (5)	30.8 (5)	36.4 (5)	24.7 (5)	-
Code							
HumanEval	35.4 (0)	34.8 (0)	12.8 (0)	51.2 (0)	57.9 (0)	29.3 (0)	26.2 (0)
HumanEval-FIM	73.8 (2)	73.3 (2)	26.9 (2)	-	-	-	-
MBPP	40.0 (4)	48.8 (4)	23.2 (4)	64.2 (0)	74.9 (0)	51.1 (0)	39.0 (3)
Chinese							
CMMLU	69.9 (5)	50.7 (5)	32.5 (5)	83.9 (5)	-	-	47.2 (5)
C-Eval	70.5 (5)	51.7 (5)	34.0 (5)	83.2 (5)	-	-	45.0 (5)

Pre-trained Benchmark Results (In-Context Learning)

- Broad comparison to similarly sized LLMs on general, math, code, Chinese tasks.
- Key message: LLaDA 8B Base is competitive with strong AR baselines; advantages in some domains.

Post-trained (SFT) Benchmark Results

Table 2: **Benchmark Results of Post-trained LLMs.** LLaDA only employs an SFT procedure, while other models have extra reinforcement learning (RL) alignment. * indicates models are evaluated under the same protocol, detailed in Appendix B.6. Results indicated by [†] and [¶] are sourced from Yang et al. [26] and Bi et al. [32] respectively. The numbers in parentheses represent the number of shots used for in-context learning. “-” indicates unknown data.

	LLaDA 8B*	LLaMA3 8B*	LLaMA2 7B*	Qwen2 7B [†]	Qwen2.5 7B [†]	Gemma2 9B [†]	Deepseek 7B [¶]
Model	Diffusion	AR	AR	AR	AR	AR	AR
Training tokens	2.3T	15T	2T	7T	18T	8T	2T
Post-training	SFT	SFT+RL	SFT+RL	SFT+RL	SFT+RL	SFT+RL	SFT+RL
Alignment pairs	4.5M	-	-	0.5M + -	1M + 0.15M	-	1.5M + -
General Tasks							
MMLU	65.5 (5)	68.4 (5)	44.1 (5)	-	-	-	49.4 (0)
MMLU-pro	37.0 (0)	41.9 (0)	4.6 (0)	44.1 (5)	56.3 (5)	52.1 (5)	-
Hellaswag	74.6 (0)	75.5 (0)	51.5 (0)	-	-	-	68.5 (-)
ARC-C	88.5 (0)	82.4 (0)	57.3 (0)	-	-	-	49.4 (-)
Mathematics & Science							
GSM8K	69.4 (4)	78.3 (4)	29.0 (4)	85.7 (0)	91.6 (0)	76.7 (0)	63.0 (0)
Math	31.9 (0)	29.6 (0)	3.8 (0)	52.9 (0)	75.5 (0)	44.3 (0)	15.8 (0)
GPQA	33.3 (5)	31.9 (5)	28.4 (5)	34.3 (0)	36.4 (0)	32.8 (0)	-
Code							
HumanEval	49.4 (0)	59.8 (0)	16.5 (0)	79.9 (0)	84.8 (0)	68.9 (0)	48.2 (-)
MBPP	41.0 (4)	57.6 (4)	20.6 (4)	67.2 (0)	79.2 (0)	74.9 (0)	35.2 (-)



Post-trained (SFT) Benchmark Results

- After SFT, LLaDA improves instruction-following and task performance under conditional generation.
- Compare against instruct-tuned AR models under reported evaluation protocols.

Details and Ablation on Sampling Strategies

- Flexible Sampling Strategies

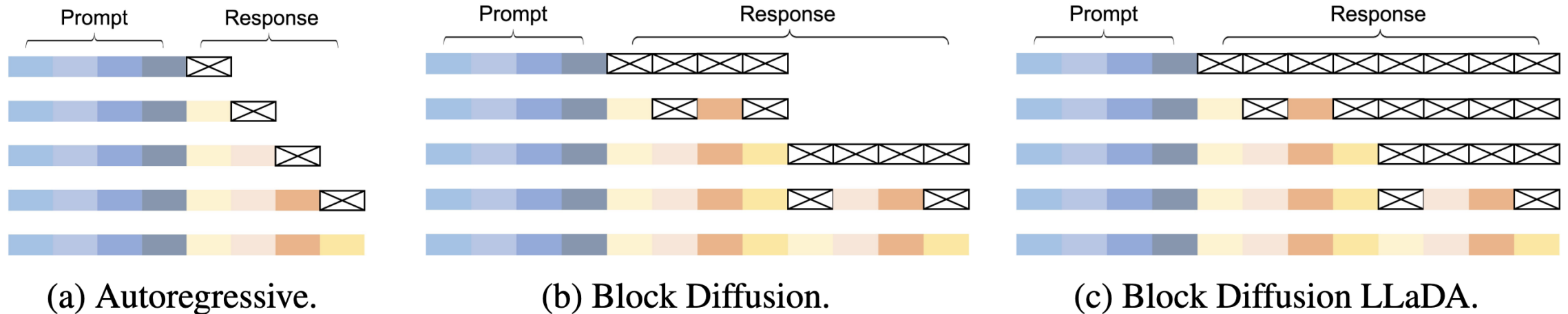


Figure 4: **Flexible Sampling Strategies Supported by LLaDA.** Colored squares depict non-masked tokens, while squares marked with $\times$ denote masked tokens. In this illustration, the block length for both block diffusion and block diffusion LLaDA sampling is 4.

Details and Ablation on Sampling Strategies

- Results

Table 7: **Ablation on Sampling Strategies for LLaDA 8B Base.** L' is the block length. Pure diffusion sampling achieves the best overall performance.

		BBH	GSM8K	Math	HumanEval	MBPP
Autoregressive		38.1	63.1	23.6	18.3	33.4
Block Difusion	$L' = 2$	37.3	62.6	25.2	14.6	33.6
	$L' = 4$	40.0	65.7	26.6	15.9	36.0
	$L' = 8$	42.0	68.2	27.7	19.5	39.2
	$L' = 32$	45.7	68.6	29.7	29.9	37.4
Block Diffusion LLaDA	$L' = 2$	48.0	70.0	30.8	26.2	40.0
	$L' = 4$	48.5	70.3	31.3	27.4	38.8
	$L' = 8$	48.6	70.2	30.9	31.1	39.0
	$L' = 32$	48.3	70.3	31.2	32.3	40.0
Pure Diffusion		49.7	70.3	31.4	35.4	40.0

Details and Ablation on Sampling Strategies

- Results

Table 8: **Ablation on Sampling Strategies for LLaDA 8B Instruct.** The block length is set to 32 for efficiency. Pure diffusion sampling achieves the best overall performance.

	GSM8K	Math	HumanEval	MBPP	GPQA	MMLU-Pro	ARC-C
Autoregressive	0	9.5	0	0	0	0	84.4
Block Diffusion	24.6	23.5	17.1	21.2	29.3	32.5	88.1
Block Difusion LLaDA	77.5	42.2	46.3	34.2	31.3	34.8	85.4
Pure Diffusion	69.4	31.9	49.4	41.0	33.3	37.0	88.5

Table 9: **Analysis on Random and Low-confidence Remasking Strategies.** The low-confidence remasking consistently outperforms the random remasking.

Length	BBH	GSM8K	Math	HumanEval	MBPP
Random Remasking	32.1	21.3	9.2	11.6	21.0
Low-confidence Remasking	45.0	70.0	30.3	32.9	40.2

Analysis of Sampling Efficiency

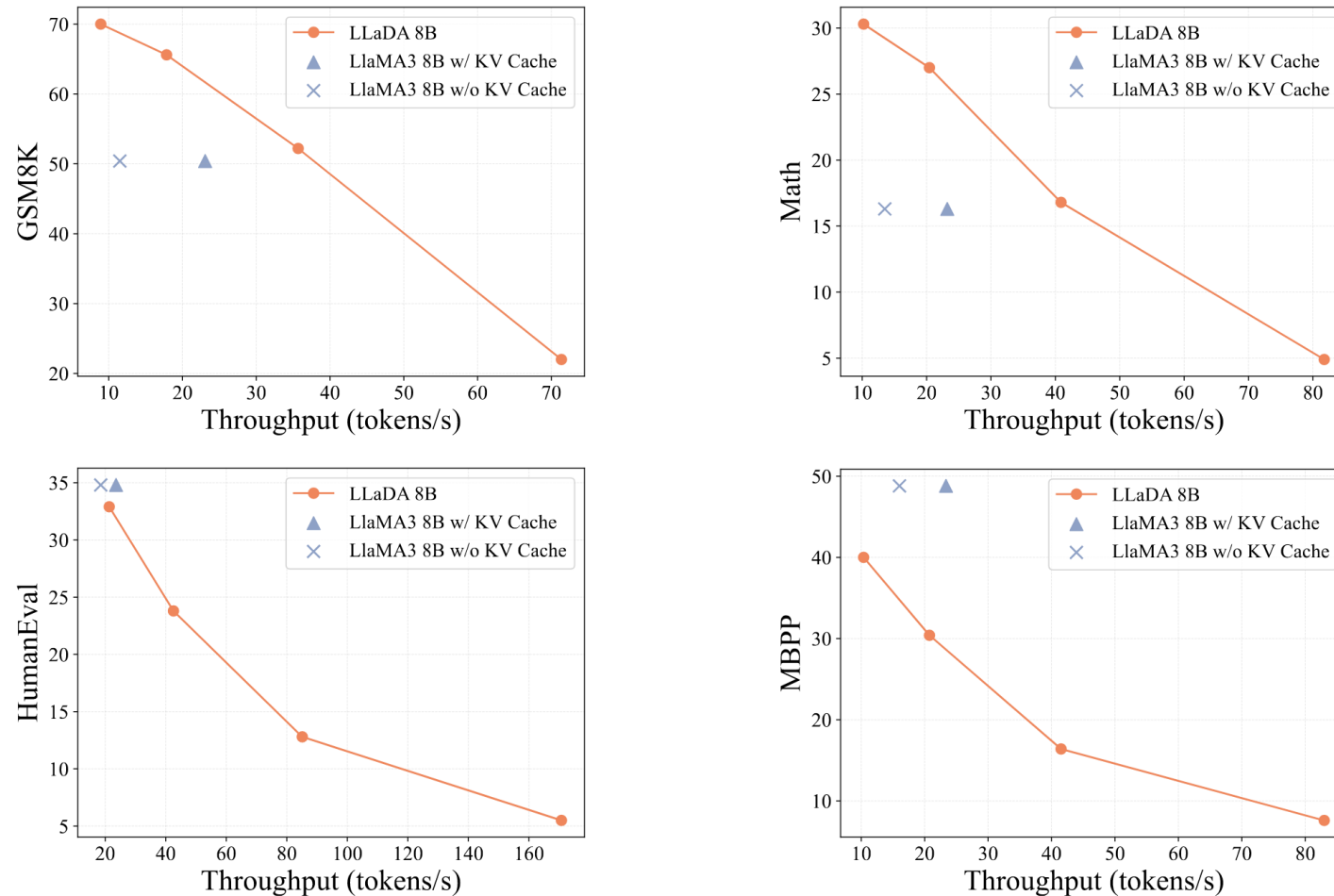


Figure 5: **Analysis of Sampling Efficiency.** The generation length for LLaDA is set to 256, with sampling steps set to 32, 64, 128, and 256 across the figures. This corresponds to decoding 8, 4, 2, and 1 token(s) per forward pass, respectively. LLaDA enables a flexible trade-off between generation

Ablation on CFG

- To ensure a fair comparison with ARMs, we do not apply CFG to LLaDA in the main text.
- However, LLaDA is fully compatible with CFG and consistently exhibits improved performance when it is applied.

Table 6: **Ablation on CFG.** CFG consistently improves the performance of LLaDA.

	ARC-C	Hellaswag	TruthfulQA	WinoGrande	GPQA	PIQA
w/o CFG	45.9	70.5	46.1	74.8	25.2	73.6
w/ CFG	47.9	72.5	46.4	74.8	26.1	74.4

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Reversal Reasoning

- Motivation and Setup
 - Evaluate “reversal curse”: forward next-line vs reversal previous-line generation.
 - Dataset: Chinese poem sentence pairs; zero-shot generation without additional tuning.
 - Key claim: LLaDA treats tokens more uniformly (less left-to-right inductive bias).
- Main Results
 - Show forward vs reversal performance; LLaDA narrows the gap and improves reversal accuracy.
 - Talking point: strong AR models excel forward but drop sharply on reversal; LLaDA is more balanced.

Table 4: **Comparison on the Poem Completion task.**

	Forward	Reversal
GPT-4o (2024-08-06)	82.7	34.3
Qwen2.5-7B Instruct	75.9	38.0
LLaDA-8B Instruct	51.8	45.6

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Instruction Following Case Studies (Qualitative)

Table 3: **Visualization of the Sampling Process and a Generated Multi-round Dialogue.** In the response of LLaDA, darker colors indicate tokens predicted in the later stages of sampling, while lighter colors correspond to earlier predictions.

Sampling Process	
User	Lily can run 12 kilometers per hour for 4 hours. After that, she runs 6 kilometers per hour. How many kilometers can she run in 8 hours?
LLaDA	In4hours, Lily runs 12 * 4 = 48 kilometers . After 4 hours she runs 6 * 4 = 24 kilometers . In total she runs 48 + 24 = 72 kilometers in 8 hours . Result 72
Multi-round Dialogue Case	
User	Could you please share with me what the initial two lines of the renowned poem ‘The Road Not Taken’?
LLaDA	Certainly! The initial two lines of the renowned poem “The Road Not Taken” by Robert Frost are: “Two roads diverged in a yellow wood, And sorry I could not travel both.” These lines set the stage for the poem, introducing the idea of a speaker facing a choice between two paths.
User	Please help me translate into Chinese.
LLaDA	Sure! The first two lines of “The Road Not Taken” by Robert Frost can be translated into Chinese as: “两条路分岔在黄色的树林中，遗憾我不能同时走”
User	Please translate into German.
LLaDA	Sure! The first two lines of “The Road Not Taken” by Robert Frost can be translated into German as: “Zwei Wege trennten sich im gelben Wald, und leider konnte ich nicht beide tun.”
User	Please also write a poem about life choices. I require 5 sentences, and the first word of each sentence starts with C as the first letter.
LLaDA	Certainly! Here’s a poem about life choices, with each sentence starting with the letter “C”: Choices cascade through life’s journey, Creating connections with each decision, Careful consideration leads to better paths, Courageous choices shape our destiny, Choosing wisely guides us through the years.

Instruction Following Case Studies (Qualitative)

Table 13: Visualization of the Block Diffusion LLaDA Sampling Process. In the response of LLaDA, darker colors indicate tokens predicted in the later stages of sampling, while lighter colors correspond to earlier predictions.

User
LLaDA

What are the benefits of regular exercise for physical and mental health?

Regular exercise can improve physical health by strengthening muscles and bones , improving cardiovascular health , and reducing the risk of chronic diseases such as diabetes and heart disease . It can also improve mental health by reducing stress and anxiety , improving mood , and enhancing cognitive function . Additionally , exercise can improve sleep quality and increase energy levels .

Instruction Following Case Studies (Qualitative)

Table 14: **Visualization of the Multi-turn Dialogue.** We employ random remasking strategy. In the response of LLaDA, darker colors indicate tokens predicted in the later stages of sampling, while lighter colors correspond to earlier predictions.

User	I want to have a pet, but I don't know whether to get a cat or a dog. Do you have any suggestions?
LLaDA	It really depends on your preferences , lifestyle , and the type of pet you want . If you want an independent pet and have a busy schedule , a cat is a good choice . If you prefer a companion and have a more active lifestyle , a dog might be a better fit . Have you considered your options ?
User	I'm very busy at work, so it seems like a cat would be a more suitable option. Do you have any suggestions for getting a cat?
LLaDA	Sure ! First , make sure you have enough space in your home and provide food , water , and toys for your cat . Also , research different breeds and personalities , and choose cats that are suitable for your lifestyle . Lastly , make sure you have the necessary vaccinations and insurance in place before you get a pet .

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Conclusion and Discussion

- Core takeaway: diffusion language models can exhibit LLM-like capabilities at scale.
- Highlights: competitive benchmark performance, instruction-following after SFT, reversal robustness.
- Broader impacts: similar societal concerns as LLMs (misuse, bias, environmental cost).
- Limitation
 - The generation length is a user-specified hyperparameter.
 - No specialized attention mechanisms or position embeddings were designed for LLaDA, nor were any system-level architectural optimizations such as KV cache applied.

Thank You!